Lecture Notes

Lecture 8

Orthonormal Bases, Gram–Schmidt, and Riesz

In this lecture we meet the *Gram–Schmidt process* and show that every finite-dimensional inner product space has an *orthonormal basis*. We will apply orthonormal bases to two problems. First, orthogonal projection will find the point of a subspace closest to a given vector (the *best approximation*). Second, the *Riesz representation theorem* will show that every linear functional is a bra. These are the daily tools of quantum mechanics: a state is expanded in an eigenbasis, and the probabilities of measurement outcomes are squared moduli of inner products.

Learning objectives

By the end of this lecture you will be able to:

- Expand vectors in an orthonormal basis, with coordinates as inner products.
- Build an orthonormal basis by the Gram–Schmidt process.
- Compute orthogonal projections.
- Solve best-approximation and least-squares problems.
- Link the Riesz representation theorem to the bra–ket correspondence.

1 Orthonormal bases

Definition 8.1 (Orthonormal list and basis). A list $e_1, \dots, e_m$ is *orthonormal* if

$$\langle e_j, e_k \rangle = \delta_{jk},$$

i.e. each vector is a unit vector and distinct vectors are orthogonal. An orthonormal list that is also a basis is an *orthonormal basis*.

Lemma 8.2 (Orthonormal lists are independent). Every orthonormal list is linearly independent.

Proof. Let $\sum_k c_k e_k = 0$. Pairing this equation with e_j shows that every coefficient vanishes:

$$0 = \langle e_j, \sum_k c_k e_k \rangle = \sum_k c_k \delta_{jk} = c_j. \qquad \blacksquare$$

In an orthonormal basis, finding coordinates requires no linear system: the coordinates are inner products.

Theorem 8.3 (Expansion in an orthonormal basis). If $e_1, \dots, e_n$ is an orthonormal basis of V, then for every $v \in V$

$$v = \sum_{k=1}^{n} \langle e_k, v \rangle \, e_k, \qquad \|v\|^2 = \sum_{k=1}^{n} |\langle e_k, v \rangle|^2.$$

The second identity is *Parseval's relation*.

Proof. The e_k form a basis, so $v = \sum_k c_k e_k$. Pairing with e_j and using orthonormality gives $\langle e_j, v\rangle = \sum_k c_k \delta_{jk} = c_j$, which is the first formula. The second follows from

$$\|v\|^2 = \langle v, v\rangle = \sum_{j,k} \overline{c_j} c_k \langle e_j, e_k\rangle = \sum_k |c_k|^2. \qquad \blacksquare$$

Corollary 8.4 (Inner products from coordinates). In an orthonormal basis

$$\langle u, v\rangle = \sum_k \overline{\langle e_k, u\rangle}\, \langle e_k, v\rangle.$$

Proof. Expand $u = \sum_j \langle e_j, u\rangle e_j$ and $v = \sum_k \langle e_k, v\rangle e_k$. Then

$$\langle u, v\rangle = \sum_{j,k} \overline{\langle e_j, u\rangle}\langle e_k, v\rangle\langle e_j, e_k\rangle = \sum_k \overline{\langle e_k, u\rangle}\langle e_k, v\rangle. \qquad \blacksquare$$

So in an orthonormal basis the inner product is always the standard sum of coordinates

$$\sum_k \overline{a_k} b_k.$$

Remark 8.5 (Completeness relation). In Dirac notation the expansion reads

$$|v\rangle = \sum_k |e_k\rangle\langle e_k, v\rangle = \Big(\sum_k |e_k\rangle\langle e_k|\Big)|v\rangle,$$

so an orthonormal basis satisfies the *completeness relation* $\sum_k |e_k\rangle\langle e_k| = I$. In quantum mechanics this resolution of the identity is routinely inserted between two symbols: that is how any expression is written in coordinates of a chosen basis.

Example 8.6 (Three orthonormal bases).

(1) The standard basis of $\mathbb{C}^n$ is orthonormal.

(2) So are the qubit vectors $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ of Lecture 7.

(3) On the space of trigonometric polynomials of degree at most N, the functions

$$\frac{1}{\sqrt{2\pi}}, \qquad \frac{1}{\sqrt{\pi}}\cos kx, \qquad \frac{1}{\sqrt{\pi}}\sin kx, \qquad k = 1, \dots, N,$$

are orthonormal for the inner product $\langle f, g\rangle = \int_{-\pi}^{\pi} \overline{f}\, g\, dx$. This basis underlies Fourier analysis. ◁

Example 8.7 (Coordinates as inner products). Expand $v = (3, 4) \in \mathbb{R}^2$ in the orthonormal basis

$$e_1 = \tfrac{1}{\sqrt{2}}(1, 1) \quad \text{and} \quad e_2 = \tfrac{1}{\sqrt{2}}(1, -1).$$

The coordinates are $\langle e_1, v\rangle = \frac{7}{\sqrt{2}}$ and $\langle e_2, v\rangle = -\frac{1}{\sqrt{2}}$, so

$$v = \tfrac{7}{\sqrt{2}}\, e_1 - \tfrac{1}{\sqrt{2}}\, e_2.$$

Parseval checks: $\frac{49}{2} + \frac{1}{2} = 25 = \|v\|^2$. No linear system was solved. ◁

Example 8.8 (A Fourier coefficient). In the Fourier basis of Example 8.6, the coordinate of a function f of that space along the basis function $\frac{1}{\sqrt{\pi}}\sin kx$ is

$$\frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} \sin(kx) f(x)\, dx = \sqrt{\pi}\, b_k,$$

where $b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)\, dx$ is the kth Fourier sine coefficient. Parseval's relation says that the "energy" $\|f\|^2$ equals the sum of the squared moduli of all the coordinates—the foundation of spectral signal analysis. ◁

2 The Gram–Schmidt process

Theorem 8.9 (Gram–Schmidt). Every finite-dimensional inner product space has an orthonormal basis. Concretely, from any basis $v_1, \dots, v_n$ the formulas

$$e_1 = \frac{v_1}{\|v_1\|}, \qquad e_k = \frac{w_k}{\|w_k\|}, \qquad w_k = v_k - \sum_{j<k} \langle e_j, v_k \rangle \, e_j,$$

build an orthonormal basis $e_1, \dots, e_n$ with $\operatorname{span}(e_1, \dots, e_k) = \operatorname{span}(v_1, \dots, v_k)$ for every k.

Proof. We prove the theorem by induction on k.

1. For $k = 1$, the vector v_1 is nonzero, because it belongs to a basis. Hence $e_1 = v_1/\|v_1\|$ is a unit vector and $\operatorname{span}(e_1) = \operatorname{span}(v_1)$.
2. Assume that $e_1, \dots, e_{k-1}$ are already orthonormal and span the same subspace as the vectors $v_1, \dots, v_{k-1}$.
3. *The vector w_k is nonzero.* If $w_k = 0$, the vector v_k would be a linear combination of $e_1, \dots, e_{k-1}$, and hence also of $v_1, \dots, v_{k-1}$. This contradicts the linear independence of $v_1, \dots, v_n$. Hence $e_k = w_k/\|w_k\|$ is defined and is a unit vector.
4. *Orthogonality.* For $i < k$,

$$\langle e_i, w_k \rangle = \langle e_i, v_k \rangle - \sum_{j<k} \langle e_j, v_k \rangle \langle e_i, e_j \rangle = \langle e_i, v_k \rangle - \langle e_i, v_k \rangle = 0,$$

   so e_k is orthogonal to all the earlier vectors.
5. *Spans.* The vectors v_k and $\|w_k\| \, e_k$ differ only by a linear combination of $e_1, \dots, e_{k-1}$. Hence $\operatorname{span}(e_1, \dots, e_k) = \operatorname{span}(e_1, \dots, e_{k-1}, v_k)$, and by the induction hypothesis this subspace equals $\operatorname{span}(v_1, \dots, v_k)$.
6. For $k = n$ we have n orthonormal vectors. By Lemma 8.2 they are linearly independent, hence a basis. ∎

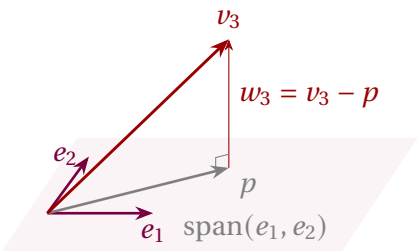


Figure 1: The third Gram–Schmidt step. Subtract from v_3 the vector $p = \langle e_1, v_3 \rangle e_1 + \langle e_2, v_3 \rangle e_2$ of the plane $\operatorname{span}(e_1, e_2)$. The remainder $w_3 = v_3 - p$ is orthogonal to the plane; normalizing it gives e_3.

Example 8.10 (Gram–Schmidt in $\mathbb{R}^3$). Apply the process to

$$v_1 = (1, 1, 0), \qquad v_2 = (2, 0, 1), \qquad v_3 = (1, -2, -2).$$

First $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$. Since $\langle e_1, v_2 \rangle = \sqrt{2}$,

$$v_2 - \langle e_1, v_2 \rangle e_1 = (2, 0, 1) - (1, 1, 0) = (1, -1, 1), \qquad e_2 = \tfrac{1}{\sqrt{3}}(1, -1, 1).$$

Finally $\langle e_1, v_3\rangle = -\frac{1}{\sqrt{2}}$ and $\langle e_2, v_3\rangle = \frac{1}{\sqrt{3}}$, so

$$v_3 - \langle e_1, v_3\rangle e_1 - \langle e_2, v_3\rangle e_2 = (1,-2,-2) + \tfrac{1}{2}(1,1,0) - \tfrac{1}{3}(1,-1,1) = \tfrac{7}{6}(1,-1,-2),$$

and $e_3 = \frac{1}{\sqrt{6}}(1,-1,-2)$. Check: $\langle e_i, e_j\rangle = \delta_{ij}$, e.g. $\langle e_2, e_3\rangle = \frac{1}{\sqrt{18}}(1+1-2) = 0$. This is the quickest test of a Gram–Schmidt computation. ◁

Example 8.11 (Gram–Schmidt over $\mathbb{C}$). Orthonormalize $v_1 = (1, i)$ and $v_2 = (1, 0)$ in $\mathbb{C}^2$. First $e_1 = \frac{1}{\sqrt{2}}(1, i)$. Then $\langle e_1, v_2\rangle = \frac{1}{\sqrt{2}}\overline{1}\cdot 1 = \frac{1}{\sqrt{2}}$, so

$$v_2 - \langle e_1, v_2\rangle e_1 = (1,0) - \tfrac{1}{2}(1,i) = \tfrac{1}{2}(1,-i).$$

Normalizing gives $e_2 = \frac{1}{\sqrt{2}}(1,-i)$. Check: $\langle e_1, e_2\rangle = \frac{1}{2}(1+i^2) = 0$. Over $\mathbb{C}$, do not forget to conjugate the coordinates of the first vector. ◁

Example 8.12 (Legendre polynomials). On $\mathcal{P}_2$ with the inner product $\langle p, q\rangle = \int_{-1}^{1} \overline{p}\, q\, dx$, orthonormalize $1, x, x^2$. The polynomial 1 has squared norm $\int_{-1}^{1} 1\, dx = 2$, so normalizing it gives p_0. The polynomial x is already orthogonal to p_0, because $\langle p_0, x\rangle = 0$, so we only normalize it and get p_1. The third step gives the residual $x^2 - \langle p_0, x^2\rangle p_0 - \langle p_1, x^2\rangle p_1 = x^2 - \frac{1}{3}$; normalizing it gives p_2. So the orthonormal polynomials are

$$p_0 = \tfrac{1}{\sqrt{2}}, \qquad p_1 = \sqrt{\tfrac{3}{2}}\, x, \qquad p_2 = \sqrt{\tfrac{5}{8}}\,(3x^2 - 1).$$

Up to the factors $\sqrt{(2k+1)/2}$, these are the first *Legendre polynomials* 1, x, $\frac{1}{2}(3x^2-1)$, ubiquitous in electrostatics and the quantum theory of angular momentum. ◁

Example 8.13 (Orthonormal basis of a plane). The plane

$$U = \{(x, y, z) : x - y - 2z = 0\} \subset \mathbb{R}^3$$

contains the linearly independent vectors $v_1 = (2, 0, 1)$ and $v_2 = (3, 1, 1)$. Gram–Schmidt gives $e_1 = \frac{1}{\sqrt{5}}(2, 0, 1)$. Since $\langle e_1, v_2\rangle = \frac{7}{\sqrt{5}}$,

$$v_2 - \langle e_1, v_2\rangle e_1 = (3,1,1) - \tfrac{7}{5}(2,0,1) = \tfrac{1}{5}(1,5,-2),$$

and $e_2 = \frac{1}{\sqrt{30}}(1, 5, -2)$. These orthonormal vectors form a basis of the plane U. ◁

> **Corollary 8.14 (Extending to an orthonormal basis).** Every orthonormal list in a finite-dimensional V extends to an orthonormal basis of V.

Proof. The orthonormal list is linearly independent (Lemma 8.2), so it extends to a basis (Lecture 2). Run Gram–Schmidt on that basis. The already-orthonormal vectors are unchanged, and the rest become an orthonormal completion. ∎

By Gram–Schmidt, every n-dimensional complex inner product space has an orthonormal basis, and by Corollary 8.4 the coordinates in it turn the inner product into the standard one. So $\mathbb{C}^n$ with its standard inner product is the model of every such space: each is *isometric* to it, i.e. isomorphic by a map that preserves not only the linear structure but also inner products, and hence norms.

3 Orthogonal projection and best approximation

Definition 8.15 (Orthogonal projection). Let U be a subspace of V with orthonormal basis $e_1, \dots, e_m$. The *orthogonal projection* onto U is the operator

$$P_U v = \sum_{k=1}^{m} \langle e_k, v \rangle \, e_k.$$

Theorem 8.16 (Orthogonal decomposition). For each $v \in V$, $v - P_U v \in U^\perp$, and $V = U \oplus U^\perp$. Thus P_U is the projection onto U along $U^\perp$.

Proof. For each i,

$$\langle e_i, v - P_U v \rangle = \langle e_i, v \rangle - \sum_k \langle e_k, v \rangle \langle e_i, e_k \rangle = \langle e_i, v \rangle - \langle e_i, v \rangle = 0.$$

So $v - P_U v$ is orthogonal to every e_i, and hence to U. The identity $v = P_U v + (v - P_U v)$ writes v as the sum of a vector of U and a vector of $U^\perp$. Moreover $U \cap U^\perp = \{0\}$, since a vector orthogonal to itself is zero. Hence the sum is direct. ∎

The formula uses a basis, but the operator does not depend on it. By Theorem 8.16, every vector v is uniquely the sum of a vector of U and a vector of $U^\perp$, and $P_U v$ is exactly the first summand. No basis appears in this description, so any other orthonormal basis of U yields the same operator P_U. The vector p of Figure 1 is $P_U v_3$ for $U = \operatorname{span}(e_1, e_2)$: each Gram–Schmidt step subtracts the projection onto the span of the vectors already built.

Proposition 8.17 (Dimension of the orthogonal complement). For a subspace U of a finite-dimensional V, $\dim U + \dim U^\perp = \dim V$, and $(U^\perp)^\perp = U$.

Proof. Take an orthonormal basis $e_1, \dots, e_m$ of U and extend it to an orthonormal basis $e_1, \dots, e_n$ of V (Corollary 8.14). The extra vectors $e_{m+1}, \dots, e_n$ lie in $U^\perp$. They also span $U^\perp$, because any $v \in U^\perp$ has zero coordinates along $e_1, \dots, e_m$. So $\dim U^\perp = n - m$.

For the second claim, the inclusion $U \subseteq (U^\perp)^\perp$ is immediate from the definition: every $u \in U$ is orthogonal to every vector of $U^\perp$. Applying the dimension identity to $U^\perp$ gives $\dim (U^\perp)^\perp = n - (n - m) = m = \dim U$. A subspace contained in another of the same finite dimension equals it, so $(U^\perp)^\perp = U$. ∎

Example 8.18 (Computing an orthogonal complement). Let $U = \operatorname{span}\big((1, 2, 3, -4)\big) \subset \mathbb{R}^4$. The orthogonal complement $U^\perp$ is the solution set of the equation

$$\langle (1, 2, 3, -4), v \rangle = v_1 + 2v_2 + 3v_3 - 4v_4 = 0.$$

Taking v_2, v_3 and v_4 as the free variables and setting one of them to 1 and the others to 0, we read off the basis vectors and write down the orthogonal complement:

$$U^\perp = \operatorname{span}\big((-2, 1, 0, 0),\ (-3, 0, 1, 0),\ (4, 0, 0, 1)\big).$$

This is a three-dimensional hyperplane, as $\dim U + \dim U^\perp = 1 + 3 = 4$ requires. ◁

Theorem 8.19 (Best approximation). Let $v \in V$. For every $u \in U$,

$$\|v - P_U v\| \le \|v - u\|,$$

with equality only when $u = P_U v$.

Proof. Write $v - u = (v - P_U v) + (P_U v - u)$. The first piece lies in $U^\perp$ and the second in U, so they are orthogonal. By the Pythagorean theorem

$$\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \ge \|v - P_U v\|^2,$$

with equality iff $\|P_U v - u\| = 0$, i.e. $u = P_U v$. ∎

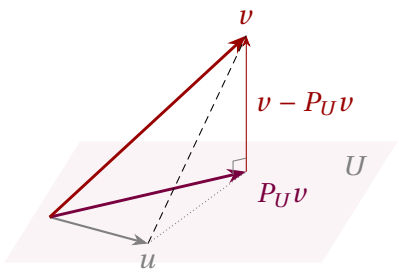


Figure 2: Best approximation. Of all vectors u of U, the orthogonal projection $P_U v$ is closest to v, because the error $v - P_U v$ is perpendicular to U.

Example 8.20 (Projection onto a plane). Take the orthonormal vectors e_1, e_2 of Example 8.13. The projection onto that plane is $P_U = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|$. For $b = (1, 0, 0)$,

$$P_U b = \langle e_1, b\rangle e_1 + \langle e_2, b\rangle e_2 = \tfrac{2}{5}(2, 0, 1) + \tfrac{1}{30}(1, 5, -2) = \tfrac{1}{6}(5, 1, 2).$$

The residual $b - P_U b = \frac{1}{6}(1, -1, -2)$ is parallel to the plane's normal $(1, -1, -2)$, as the orthogonal decomposition demands. ◁

Corollary 8.21 (Bessel's inequality). For any orthonormal vectors $e_1, \dots, e_m$ and any vector v,

$$\sum_{k=1}^{m} |\langle e_k, v\rangle|^2 \le \|v\|^2.$$

Proof. Take $U = \operatorname{span}(e_1, \dots, e_m)$. Parseval on U gives $\|P_U v\|^2 = \sum_k |\langle e_k, v\rangle|^2$. Moreover $v - P_U v \perp P_U v$, so by the Pythagorean theorem $\|P_U v\| \le \|v\|$. ∎

Example 8.22 (Least squares and the normal equations). Orthogonal projection also solves the *least-squares* problem. Suppose the system $Ax = b$ has more equations than unknowns and is inconsistent. We then look for x making $\|Ax - b\|$ as small as possible. The vectors Ax form the subspace $\operatorname{im} A$, so by the best-approximation theorem Ax must be the orthogonal projection of b onto $\operatorname{im} A$. Hence the residual $b - Ax$ is orthogonal to every vector Ay:

$$\langle Ay, b - Ax\rangle = 0 \qquad \text{for all } y.$$

For the standard inner product, $\langle Ay, r\rangle = (Ay)^\dagger r = y^\dagger A^\dagger r = \langle y, A^\dagger r\rangle$, so $\langle y, A^\dagger(b - Ax)\rangle = 0$ for all y. Taking $y = A^\dagger(b - Ax)$ gives $A^\dagger(b - Ax) = 0$, i.e. the *normal equations*

$$A^\dagger A x = A^\dagger b.$$

Conversely, if x solves them, the same computation read backwards gives $b - Ax \perp \operatorname{im} A$, so Ax is the projection of b onto $\operatorname{im} A$ by the uniqueness of the orthogonal decomposition. So the solutions of the normal equations are exactly the minimizers of $\|Ax - b\|$. ◁

Example 8.23 (Least-squares line fit). We seek the line $y = c_0 + c_1 x$ that best fits the points $(0,1), (1,3), (2,4)$. No line passes through all three, so the system $Ac = y$ with

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$$

is inconsistent. Multiplying out,

$$A^{\dagger}A = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix} \quad \text{and} \quad A^{\dagger}y = \begin{pmatrix} 8 \\ 11 \end{pmatrix}.$$

We solve the normal equations $A^{\dagger}Ac = A^{\dagger}y$ by reducing the augmented matrix:

$$\left[\begin{array}{cc|c} 3 & 3 & 8 \\ 3 & 5 & 11 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} \textcircled{1} & 0 & \frac{7}{6} \\ 0 & \textcircled{1} & \frac{3}{2} \end{array}\right] \Longrightarrow c_0 = \tfrac{7}{6}, \quad c_1 = \tfrac{3}{2}.$$

The line $y = \frac{7}{6} + \frac{3}{2}x$ minimizes the sum of squared vertical errors. Its values at $x = 0, 1, 2$ form the orthogonal projection of y onto $\operatorname{im} A$. ◁

Example 8.24 (Best polynomial approximation of $\sin$). Project $v(x) = \sin x$ onto the subspace of polynomials of degree at most 5 in $\mathscr{C}[-\pi, \pi]$. To do so, run Gram–Schmidt on $1, x, \dots, x^5$ and then apply P_U. The result is the best approximation in the mean-square sense:

$$u(x) \approx 0.98786\, x - 0.15527\, x^3 + 0.00564\, x^5.$$

At $\pm\pi$ it is far more accurate than the Taylor polynomial $x - \frac{x^3}{3!} + \frac{x^5}{5!}$. Orthogonal projection minimizes the mean-square error over the whole interval, whereas the Taylor polynomial matches the derivatives at a single point. ◁

4 The Riesz representation theorem

Recall that a *linear functional* on V is a linear map $\varphi : V \to \mathbb{F}$ to the scalars, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ (Lecture 7). In an inner product space every linear functional is the inner product with a fixed vector.

Theorem 8.25 (Riesz representation). Let V be finite-dimensional. For every linear functional $\varphi : V \to \mathbb{F}$ there is a unique $w \in V$ such that

$$\varphi(v) = \langle w, v \rangle \qquad \text{for all } v \in V.$$

Proof. Fix an orthonormal basis $e_1, \dots, e_n$ and set $w = \sum_{k} \overline{\varphi(e_k)}\, e_k$. Then for any $v = \sum_k \langle e_k, v\rangle e_k$,

$$\langle w, v \rangle = \sum_{k} \varphi(e_k)\langle e_k, v\rangle = \varphi\Big(\sum_{k} \langle e_k, v\rangle e_k\Big) = \varphi(v).$$

The first step uses $\langle w, e_k\rangle = \varphi(e_k)$, and the second the linearity of φ. For uniqueness, if $\langle w, v\rangle = \langle w', v\rangle$ for all v, then $\langle w - w', v\rangle = 0$ for all v. Taking $v = w - w'$ gives $w = w'$. ∎

Remark 8.26 (Bra–ket correspondence). The Riesz theorem says that every linear functional φ is a *bra*: there is a unique *ket* $|w\rangle$ with $\varphi = \langle w|$. So the map $|w\rangle \mapsto \langle w|$ is a bijection. As we saw in Lecture 7, it is *conjugate-linear*:

$$a\,|w\rangle + b\,|u\rangle \mapsto \overline{a}\,\langle w| + \overline{b}\,\langle u|.$$

This is the essence of Dirac's notation.

Example 8.27 (A functional as an inner product). On $\mathbb{C}^2$ the functional $\varphi(v) = v_1 + 2iv_2$ equals the inner product $\langle w, v\rangle$ with $w = (\overline{1}, \overline{2i}) = (1, -2i)$:

$$\langle w, v\rangle = \overline{1}\,v_1 + \overline{(-2i)}\,v_2 = v_1 + 2iv_2.$$

The coordinates of w are the conjugates of the functional's coefficients. In Dirac notation $\varphi = \langle w|$. ◁

Example 8.28 (The trace as an inner product). On $M_{n\times n}(\mathbb{C})$ with the Frobenius product, the trace functional $\varphi(A) = \operatorname{tr} A$ is represented by the identity matrix:

$$\langle I, A\rangle = \operatorname{tr}(I^\dagger A) = \operatorname{tr} A,$$

so $w = I$. In Dirac notation $\operatorname{tr} = \langle I|$. ◁

Example 8.29 (Riesz on a polynomial space). Evaluation at 0, $\varphi(p) = p(0)$, is a linear functional on $\mathcal{P}_n$. Take the inner product of Example 8.12 on $[-1, 1]$. By Riesz there is a unique polynomial w such that

$$p(0) = \langle w, p\rangle \qquad \text{for all } p \in \mathcal{P}_n.$$

Such a w is called a *reproducing kernel*, because it reproduces the value of a polynomial at a point. For example, for $n = 2$ the formula from the proof of Riesz, with the Legendre polynomials p_0, p_1, p_2, gives

$$w = p_0(0)\,p_0 + p_1(0)\,p_1 + p_2(0)\,p_2 = \tfrac{1}{2} - \tfrac{5}{8}(3x^2 - 1) = \tfrac{9}{8} - \tfrac{15}{8}x^2.$$

The polynomial w depends on the inner product, and so also on the choice of interval. ◁

5 Measurement, expansion, and what comes next

In quantum mechanics the possible outcomes of a measurement usually correspond to the vectors of an orthonormal basis. Lectures 9–10 show that the eigenvectors of an observable (the operator describing a measured quantity) can form exactly such a basis.

Suppose that the eigenvalues $\lambda_1, \dots, \lambda_n$ of an observable are distinct, and that the corresponding eigenvectors $|e_1\rangle, \dots, |e_n\rangle$ form an orthonormal basis. An ideal (projective) measurement can return only one of the values λ_k. Expand a normalized state $|\psi\rangle$:

$$|\psi\rangle = \sum_k \langle e_k, \psi\rangle |e_k\rangle.$$

By the Born rule (Lecture 7), the outcome λ_k has probability $|\langle e_k, \psi\rangle|^2$, and Parseval's relation guarantees that the probabilities sum to $\|\psi\|^2 = 1$. If the outcome λ_k is obtained, the state is replaced by its orthogonal projection onto the line spanned by $|e_k\rangle$, divided by its norm; that is $|e_k\rangle$ up to a phase factor. When an eigenvalue is repeated, the state is projected onto the whole eigenspace (Lecture 10). The completeness relation $\sum_k |e_k\rangle\langle e_k| = I$ lets us insert a basis anywhere in a calculation.

Example 8.30 (A projective measurement). Let a spin be in the state

$$|\psi\rangle = \tfrac{1}{\sqrt{2}}|0\rangle + \tfrac{1}{\sqrt{2}}|1\rangle,$$

measured in the basis $|0\rangle, |1\rangle$. The measurement finds the system in the state $|0\rangle$ with probability $|\langle 0, \psi\rangle|^2 = \frac{1}{2}$ and in the state $|1\rangle$ with probability $\frac{1}{2}$. If $|0\rangle$ is found, the new spin state immediately after the measurement is the normalized projection

$$|\psi\rangle \longmapsto P_0|\psi\rangle / \|P_0|\psi\rangle\| = |0\rangle, \qquad P_0 = |0\rangle\langle 0|.$$

This update of the state is called the *Lüders rule*. Hence an immediate repeat of the measurement finds $|0\rangle$ again with probability 1. ◁

Example 8.31 (Expansion in an energy eigenbasis). Let the vectors $|E_n\rangle$ form an orthonormal eigenbasis of a time-independent energy operator (the Hamiltonian), with real eigenvalues E_n (the energies). Lectures 9–10 prove that a Hamiltonian has both properties. Then

$$|\psi\rangle = \sum_n \langle E_n, \psi\rangle |E_n\rangle.$$

In Lecture 6 we saw how the state evolves in time:

$$|\psi(t)\rangle = \sum_n e^{-iE_n t/\hbar} \langle E_n, \psi\rangle |E_n\rangle.$$

Each coordinate only acquires a phase factor $e^{-iE_n t/\hbar}$ of modulus 1. So the probabilities $|\langle E_n, \psi\rangle|^2$ do not change in time. This is conservation of energy. ◁

Remark 8.32 (Infinite-dimensional spaces). In an infinite-dimensional space an infinite orthonormal *list* need not be a basis. Parseval's relation may then fail, and only Bessel's inequality remains. Moreover, the decomposition $V = U \oplus U^\perp$ need not hold for every subspace U. Lecture 13 repairs both with completeness: of the space (Hilbert spaces), of the orthonormal set, and closedness of the subspace.

In the next lecture the Riesz theorem gives every operator T an *adjoint* $T^\dagger$, characterized by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$.

Summary of main concepts

An orthonormal basis satisfies $\langle e_j, e_k\rangle = \delta_{jk}$. In it, coordinates are inner products, $v = \sum_k \langle e_k, v\rangle e_k$, the norm obeys Parseval's relation $\|v\|^2 = \sum_k |\langle e_k, v\rangle|^2$, and $\sum_k |e_k\rangle\langle e_k| = I$. Gram–Schmidt builds such a basis from any basis and preserves the partial spans. So every finite-dimensional inner product space has an orthonormal basis, and every orthonormal list extends to one. Orthogonal projection $P_U v = \sum_k \langle e_k, v\rangle e_k$ gives the orthogonal decomposition $V = U \oplus U^\perp$ and solves the best-approximation problem; Bessel's inequality is a corollary. Riesz writes every linear functional as $\langle w, \cdot\rangle$. This is the bra–ket correspondence.

- **Orthonormal basis**—$\langle e_j, e_k\rangle = \delta_{jk}$; $v = \sum_k \langle e_k, v\rangle e_k$, Parseval's relation, $\sum_k |e_k\rangle\langle e_k| = I$.
- **Gram–Schmidt**—builds an orthonormal basis from any basis, preserving the partial spans.
- **Orthogonal projection and decomposition**—$V = U \oplus U^\perp$; $P_U v$ is the best approximation; Bessel's inequality.
- **Riesz representation**—every functional is $\langle w, \cdot\rangle$; the bra–ket correspondence.